COLEGIO DE POSTGRADUADOS

CAMPUS MONTECILLO

PSEI-ESTADÍSTICA Y CIENCIA DE DATOS

TARE 2. MODELOS LINEALES

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1. Sea $\mathbf{A} = \begin{bmatrix} 7 & -3 & 2 \\ 4 & 9 & 5 \end{bmatrix}$

```
A=matrix(c(7,-3,2,4,9,5),ncol=3, byrow = TRUE)
A
> A
     [,1] [,2] [,3]
[1,]    7   -3    2
[2,]    4    9    5
```

   (a) Encuentre A^t

```
t(A)
     [,1] [,2]
[1,]    7    4
[2,]   -3    9
[3,]    2    5
```

   (b) Encuentre $(A^t)^t = A$

```
t(t(A))
     [,1] [,2] [,3]
[1,]    7   -3    2
[2,]    4    9    5
```

   (c) Obtenga $A^tA = AA^t$

```
> t(A)%*%A
     [,1] [,2] [,3]
[1,]   65   15   34
[2,]   15   90   39
[3,]   34   39   29
> A%*%t(A)
     [,1] [,2]
[1,]   62   11
[2,]   11  122
```

   No son iguales.

2. Sea $\mathbf{A} = \begin{bmatrix} 2 & 4 \\ 1 & 3 \end{bmatrix}$ y $\mathbf{B} = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix}$

```
A=matrix(c(2,4,1,3), ncol = 2, byrow = TRUE)
> A
     [,1] [,2]
[1,]    2    4
[2,]    1    3
> B=matrix(c(1,3,2,-1), ncol = 2, byrow = TRUE)
> B
     [,1] [,2]
[1,]    1    3
[2,]    2   -1
```

   (a) Encuentre AB y BA

```
A%*%B
     [,1] [,2]
[1,]   10    2
[2,]    7    0
B%*%A
     [,1] [,2]
[1,]    5   13
[2,]    3    5
```

(b) Encuentre $|A|$, $|B|$ y $|AB|$ (los determinantes)

```
det(A)
[1] 2
> det(B)
[1] -7
> det(A%*%B)
[1] -14
```

(c) Obtenga $|BA|$ y compare con $|AB|$

```
> det(B%*%A)
[1] -14
> det(A%*%B)
[1] -14
```

Son iguales.

(d) Obtenga la $traza(AB)$ y compare con la $traza(BA)$

```
> trAB <- sum(diag(A%*%B))
> trAB
[1] 10
> trBA <- sum(diag(B%*%A))
> trBA
[1] 10
```

Son iguales.

(e) Obtenga los eigenvalores de AB y de BA

```
> eigen(A%*%B)$values
[1] 11.244998 -1.244998
> eigen(B%*%A)$values
[1] 11.244998 -1.244998
```

(f) Obtenga el Rango de A y de B

```
qr(A)$rank
[1] 2
> qr(B)$rank
[1] 2
```

(g) Obtenga si existe la matriz inversa de A^{-1} y de B^{-1}

```
> solve(A)
     [,1] [,2]
[1,]  1.5   -2
[2,] -0.5    1
```

```
> solve(B)
          [,1]       [,2]
[1,] 0.1428571  0.4285714
[2,] 0.2857143 -0.1428571
```

3.
$$A = \begin{bmatrix} 5 & -1 & 3 \\ -1 & 1 & 2 \\ 3 & 2 & 7 \end{bmatrix} \qquad B = \begin{bmatrix} 6 & -2 & 3 \\ 7 & 1 & 0 \\ 2 & -3 & 5 \end{bmatrix} \qquad C = \begin{bmatrix} 2 & -3 \\ -1 & 4 \\ 3 & 1 \end{bmatrix}$$

Sean
$$x = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix} \qquad y = \begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix} \qquad z = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

```
> A=matrix(c(5,-1,3,-1,1,2,3,2,7),ncol=3, byrow = TRUE)
> A
     [,1] [,2] [,3]
[1,]    5   -1    3
[2,]   -1    1    2
[3,]    3    2    7
> B=matrix(c(6,-2,3,7,1,0,2,-3,5),ncol=3, byrow = TRUE)
> B
     [,1] [,2] [,3]
[1,]    6   -2    3
[2,]    7    1    0
[3,]    2   -3    5
> C=matrix(c(2,-3,-1,4,3,1),ncol=2, byrow = TRUE)
> C
     [,1] [,2]
[1,]    2   -3
[2,]   -1    4
[3,]    3    1
> x<-matrix(c(3,-1,2),ncol=1, byrow = TRUE)
> x
     [,1]
[1,]    3
[2,]   -1
[3,]    2
> y<-matrix(c(3,2,4),ncol=1, byrow = TRUE)
> y
     [,1]
[1,]    3
[2,]    2
[3,]    4
> z<-matrix(c(2,5),ncol=1, byrow = TRUE)
> z
     [,1]
[1,]    2
[2,]    5
```

(a) Bx

```
> B %*% x
     [,1]
[1,]   26
[2,]   20
[3,]   19
```

(b) $y^t B$

```
> t(y) %*%B
     [,1] [,2] [,3]
[1,]   40  -16   29
```

(c) $x^t A x$

```
> t(x) %*%A %*%x
     [,1]
[1,]  108
```

(d) $x^t C z$

```
> t(x) %*%C %*%z
     [,1]
[1,]  -29
```

(e) $x^t x$

```
> t(x) %*%x
     [,1]
[1,]   14
```

(f) $x^t y$

```
> t(x) %*%y
     [,1]
[1,]   15
```

(g) $x x^t$

```
> x%*%t(x)
     [,1] [,2] [,3]
[1,]    9   -3    6
[2,]   -3    1   -2
[3,]    6   -2    4
```

(h) $x y^t$

```
> x%*%t(y)
     [,1] [,2] [,3]
[1,]    9    6   12
[2,]   -3   -2   -4
[3,]    6    4    8
```

(i) $B^t B$

```
> t(B) %*%B
     [,1] [,2] [,3]
[1,]   89  -11   28
[2,]  -11   14  -21
[3,]   28  -21   34
```

(j) yz^t

```
> y%*%t(z)
     [,1] [,2]
[1,]    6   15
[2,]    4   10
[3,]    8   20
```

(k) $\sqrt{y^t y}$

```
> sqrt(t(y) %*%y)
         [,1]
[1,] 5.385165
```

(l) $B^t B$

```
> t(B) %*%B
     [,1] [,2] [,3]
[1,]   89  -11   28
[2,]  -11   14  -21
[3,]   28  -21   34
```

(m) Encuentre $x + y$ y $x - y$

```
> x+y
     [,1]
[1,]    6
[2,]    1
[3,]    6
> x-y
     [,1]
[1,]    0
[2,]   -3
[3,]   -2
```

4. Escriba los sistemas de ecuaciones lineales en la forma **Ax=b**, Compare el rango de **la matriz aumentada A*** con el rango del coeficiente matriz **A** para cada uno de los siguientes sistemas de ecuaciones, clasifique los sistemas de ecuaciones. Encontrar las soluciones donde existan.

(a)

$$\begin{aligned} x_1 + 2x_2 + 3x_3 &= 6 \\ x_1 - x_2 &= 2 \\ x_1 - x_3 &= -1 \end{aligned}$$

Ax=b

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \begin{bmatrix} 6 \\ 2 \\ -1 \end{bmatrix}$$

La matriz aumentada A*

$$A* = \left[\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 1 & -1 & 0 & 2 \\ 1 & 0 & -1 & -1 \end{array}\right]$$

```
> A=matrix(c(1,1,1,2,-1,0,3,0,-1),3,3)
> b<-matrix(c(6,2,-1),3,1)
> Aum<-cbind(A,b)
> R(A)
[1] 3
> R(Aum)
[1] 3
```

El rango de las matrices A y $A*$ es el mismo 3. A es de rango completo pr lo que tiene soluciòn ùnica. Soluciòn del sistema:

```
> sol<-solve(A,b)
> sol
           [,1]
[1,]  1.1666667
[2,] -0.8333333
[3,]  2.1666667
```

(b)

$$\begin{aligned} x_1 - x_2 + 2x_3 &= 2 \\ x_1 - x_2 - x_3 &= -1 \\ 2x_1 - 2x_2 + x_3 &= 2 \end{aligned}$$

Bx=b

$$B = \begin{bmatrix} 1 & -1 & 2 \\ 1 & -1 & -1 \\ 2 & -2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}$$

La matriz aumentada B*

$$B* = \left[\begin{array}{ccc|c} 1 & -1 & 2 & 2 \\ 1 & -1 & -1 & -1 \\ 2 & -2 & 1 & 2 \end{array}\right]$$

```
> B=matrix(c(1,1,2,-1,-1,-2,2,-1,1),3,3)
> b1<-matrix(c(2,-1,2),3,1)
> AumB<-cbind(B,b1)
> R(B)
[1] 2
> R(AumB)
[1] 3
```

El rango de las matrices B y $B*$ no es el mismo. No tiene solución.

(c)

$$\begin{aligned} x_1 + x_2 + x_3 + x_4 &= 8 \\ x_1 - x_2 - x_3 - x_4 &= 6 \\ 3x_1 + x_2 + x_3 + x_4 &= 22 \end{aligned}$$

Cx=b

$$C = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & -1 \\ 3 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 8 \\ 6 \\ 22 \end{bmatrix}$$

La matriz aumentada C*

$$C* = \left[\begin{array}{rrrr|r} 1 & 1 & 1 & 1 & 8 \\ 1 & -1 & -1 & -1 & 6 \\ 3 & 1 & 1 & 1 & 22 \end{array}\right]$$

```
> C=matrix(c(1,1,3,1,-1,1,1,-1,1,1,-1,1),3,4)
> b2<-matrix(c(8,6,22),3,1)
> AumC<-cbind(C,b2)
> R(C)
[1] 2
> R(AumC)
[1] 2
```

El rango de las matrices C y $C*$ es el mismo, 2. C es de rango incompleto por lo que tiene soluciones infinitas. Una solución del sistema utilizando la inversa generalizada es:

```
> invC<-Ginv(C)
> solC<-invC%*%b2
> solC
     [,1]
[1,]    7
[2,]    1
[3,]    0
[4,]    0
```